

# SENTINEL● on-device

The on-device AI Runtime Copilot for Industrial HMI

From alarm flood → ranked decision — on the panel, offline, in seconds.

## THE PROBLEM

Operators drown in alarm floods (ISA-18.2: >10 alarms / 10 min) and data. Critical events get buried; new operators lack context; expert know-how is undocumented. Result: slower response, downtime, safety & compliance risk.

- \$10-20 B/yr lost output
- alarm floods → incidents
- expertise walks out

## OUR SOLUTION

An embedded AI copilot inside the HMI that ingests alarm, process & operator logs + live runtime context, and delivers guidance in-workflow — running fully on-device so it works on low-end panels, offline.

- ① Alarm Explanation & Contextual Guidance
- ② Root-Cause Investigation (alarm × trend × action)
- ③ Shift-Handover Auto-Brief
- ④ Operational Decision Support
- ⑤ SOP & Knowledge Retrieval (RAG)

–60%

alarm noise → ranked

–40%

response time

↑ 3×

new-operator ramp

100%

on-device / offline

## DATA SOURCES

Alarm logs · Process logs · Operator actions · Runtime context

## ON-DEVICE AI CORE

Small LLM (quantised) + local RAG vector store  
Trust layer: confidence · evidence · UNKNOWN

## COPILOT SKILLS

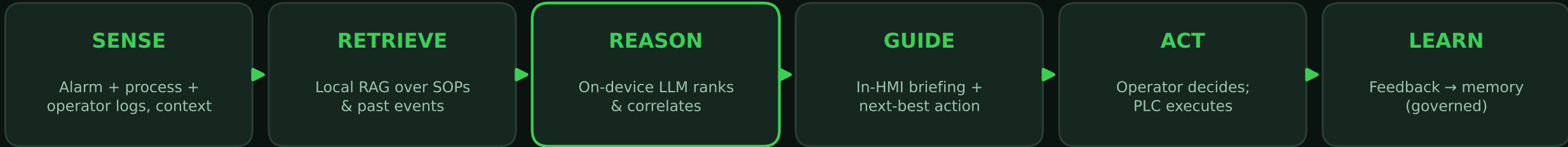
Explain · Root-cause · Handover · Decide · SOP

## HMI + OPERATOR

In-workflow guidance overlay on existing screens

► Advisory only — PLC stays authoritative. AI never controls actuators.

# How SENTINEL works — and why it wins



Closed loop · deterministic guardrails at every step · PLC remains the runtime authority.

## WHY IT WINS (differentiators)

- On-device / OFFLINE — runs on low-end HMI  
*answers PS1's low-end-hardware note others skip*
- Trust layer: cites evidence, shows UNKNOWN  
*no hallucinated plant state — safe & auditable*
- PLC-authoritative — advisory only
- In-workflow, not a side chatbot
- Captures expert knowledge → shared memory
- Beats cloud copilots at the edge

## TECH STACK

- Edge SLM (quantised) — llama.cpp / ONNX
- Local RAG — embeddings + vector store
- Runs on ARM Linux HMI / Raspberry Pi class
- Connectors: Modbus TCP · OPC UA · alarm DB
- Schneider fit: Harmony HMI · EcoStruxure  
*complements Augmented Operator Advisor*
- Open-source, low CPU/RAM footprint
- Cloud optional for fleet learning

## FEASIBILITY & WHY US

- Working HMI + AI copilot already built:  
*live 3D prototype (AssetFlow) + HMI sim*
- Trust/guardrail model already implemented
- Deterministic + LLM hybrid — demoable
- Maps 1:1 to all 5 PS-1 use cases
- Runs where the plant runs — no cloud needed
- Clear ROI: fewer floods, faster response,  
*faster onboarding, retained expertise*

“SENTINEL turns alarm floods into decisions — an on-device AI copilot that explains, investigates, and guides operators in real time, **offline on any HMI, while the PLC stays in control and every answer is backed by evidence.**”